Strict Proof: Equivalence of Spectral Locality and Zeta Function Regularity

Abstract

This paper provides the rigorous mathematical argument establishing the equivalence between three fundamental requirements of the $C(x,y)$ Paradigm: (1) Super-Exponential Locality of the Correlator Kernel $\hat{C}(x,y)$, (2) the growth restriction on the Weyl remainder term $R(\mu)$, and (3) the regularity of the spectral zeta function $\zeta_\Delta(s)$. We demonstrate that the existence of any singularity with $\mathrm{Re}(s) > 1$ in $\zeta_\Delta(s)$ is both necessary and sufficient to destroy the required macro-locality of the theory.

1. Claim: Equivalence of Spectral Regularity and Remainder Growth

Claim: The absence of singularities in the spectral zeta function $\zeta_\Delta(s)$ with $\mathrm{Re}(s) > 1$ is equivalent to the remainder term $R(\mu)$ in the Weyl law $N_\Delta(\mu) = A\mu^2 + R(\mu)$ satisfying the condition $R(\mu) = o(\mu^{1+\varepsilon})$ for any $\varepsilon > 0$.

Step A. The Link $R(\mu) \leftrightarrow$ Singularities of $\zeta_\Delta(s)$

The spectral counting function $N_\Delta(\mu)$ relates the number of eigenvalues of the Laplacian Δ less than μ. The spectral zeta function $\zeta_\Delta(s)$ is defined by the Dirichlet series over the eigenvalues λ_n of the operator Δ:

$$\boldsymbol{\zeta_\Delta(s) = \sum_n \frac{1}{\lambda_n^s}}$$

We employ the **Perron-Mellin Inversion Formula** to connect the counting function $N_\Delta(\mu)$ to the singularities of $\zeta_\Delta(s)$:

$$\mathbf{N_\Delta}(\mu) = \frac{\mathbf{1}}{\mathbf{2\pi i}} \int_{\mathbf{c-i\infty}}^{\mathbf{c+i\infty}} \boldsymbol{\zeta_\Delta}(\mathbf{s}) \frac{\mu^{\mathbf{s}}}{\mathbf{s}} \mathbf{ds}, \quad \mathbf{c > 2}$$

By shifting the contour of integration (from $c = 2 + \delta$ to the left), we express $N_\Delta(\mu)$ as the sum of residues from the poles of $\zeta_\Delta(s)$ and the integral over the new contour C':

$$\mathbf{N_\Delta}(\mu) = \sum_{\mathbf{k}} \mathrm{Res}_{\mathbf{s=s_k}} \left[\boldsymbol{\zeta_\Delta}(\mathbf{s}) \frac{\mu^{\mathbf{s}}}{\mathbf{s}} \right] + \frac{\mathbf{1}}{\mathbf{2\pi i}} \int_{\mathbf{C'}} \boldsymbol{\zeta_\Delta}(\mathbf{s}) \frac{\mu^{\mathbf{s}}}{\mathbf{s}} \mathbf{ds}$$

- **Weyl Contribution** ($s_1 = 2$): $\mathrm{Res}_{s=2}\left[\zeta_\Delta(s)\frac{\mu^s}{s}\right] = \frac{A}{2}\mu^2$ (This is the leading Weyl term for $D = 4$).
- **Next Contribution** ($s_2 = 1$): $\mathrm{Res}_{s=1}\left[\zeta_\Delta(s)\frac{\mu^s}{s}\right] = c_1\mu$ (If it exists, this pole is related to the boundary conditions and the a_2 coefficient).

The **Remainder Term** $R(\mu)$ is determined by all remaining singularities $\rho = \beta + i\gamma$:

$$\mathbf{R}(\mu) \sim \sum_{\rho} \mathrm{Res}_{\mathbf{s}=\rho} \left[\zeta_{\boldsymbol{\Delta}}(\mathbf{s}) \frac{\mu^{\mathbf{s}}}{\mathbf{s}} \right] \sim \sum_{\rho} \mathbf{c}_{\rho} \mu^{\beta} \cos(\gamma \ln \mu + \phi)$$

Conclusion: The asymptotic growth of $R(\mu)$ is governed by the singularity $\rho_{\max}$ with the largest real part $\mathrm{Re}(\rho)$. Therefore, the condition $R(\mu) = o(\mu^{1+\varepsilon})$ is rigorously equivalent to the requirement that $\mathrm{Re}(\rho_{\max}) \leq \mathbf{1}$ for all non-trivial singularities.

2. Link Between Spectral Scales

Step B. Transformation from $N_{\Delta}(\mu)$ to $N(\lambda)$

The theory introduces a new non-local scale λ through the mapping of the squared momentum μ (eigenvalue of Δ) via the Gaussian kernel:

$$\lambda = \mathbf{e}^{-\sigma^{2}\mu} \quad \Longrightarrow \quad \mu = \sigma^{-\mathbf{2}} \ln(\mathbf{1}/\lambda)$$

Substituting this into the decomposition of the counting function $N_{\Delta}(\mu)$:

$$\mathbf{N}(\lambda) = \mathbf{N}_{\boldsymbol{\Delta}}(\sigma^{-\mathbf{2}} \ln(\mathbf{1}/\lambda))$$

$$\mathbf{N}(\lambda) = \mathbf{A}(\sigma^{-\mathbf{2}} \ln(\mathbf{1}/\lambda))^{\mathbf{2}} + \mathbf{R}(\sigma^{-\mathbf{2}} \ln(\mathbf{1}/\lambda))$$

This transformation highlights that if $R(\mu)$ contains an oscillating contribution $\sim \mu^{\beta}$, the corresponding contribution to $N(\lambda)$ transforms into a logarithmic power, $\Delta N(\lambda) \sim (\ln(1/\lambda))^{\beta}$.

3. Proof of the Locality Imperative (2) $\Longleftrightarrow$ (1)

Claim: Violation of the condition $R(\mu) = o(\mu^{1+\varepsilon})$ (i.e., $\mathrm{Re}(\rho) = \beta > 1$) leads to a violation of the super-exponential locality (1) of the correlator kernel $\langle \hat{C}(x, y) \rangle$.

Step C. From Spectral Density to the Correlator $\langle \hat{C}(x, y) \rangle$

The kernel of the fundamental non-local operator $\hat{C}$ is defined as the Laplace transform of the spectral measure $dE_{\mu}(x, y)$:

$$\langle \hat{\mathbf{C}}(\mathbf{x}, \mathbf{y}) \rangle = \int_{\mathbf{0}}^{\infty} \mathbf{e}^{-\sigma^{2}\mu} \, \mathbf{dE}_{\mu}(\mathbf{x}, \mathbf{y})$$

Introducing the spectral density $\rho(x, y; \mu) = \frac{d}{d\mu} E_{\mu}(x, y)$, we analyze the integral:

$$\hat{\mathbf{C}}(\mathbf{r}) \sim \int_{\mathbf{0}}^{\infty} \rho(\mu) \mathbf{e}^{-\sigma^{2}\mu} \, \mathbf{d}\mu, \quad \mathbf{r} = \mathrm{dist}(\mathbf{x}, \mathbf{y})$$

The spectral density is decomposed into the leading and residual parts:
$\rho(x, y; \mu) = \rho_{\mathrm{Weyl}}(\mu) + \Delta\rho(x, y; \mu)$. If $R(\mu) \sim \mu^{\beta}$, then the residual density growth is $\Delta\rho(\mu) \sim \frac{d}{d\mu} R(\mu) \sim \mu^{\beta-1}$.

Asymptotic Evaluation and the Paley–Wiener Criterion

1. **Weyl Contribution ($\rho_{\text{Weyl}} \sim \mu$):** The leading contribution $\langle C(r) \rangle_{\text{Weyl}}$ is dominated by the Fourier transform of μ multiplied by $e^{-\sigma^2 \mu}$ (since $\mu \sim k^2$). This precisely yields a Gaussian kernel (in flat space):

$$\langle \mathbf{\hat{C}(r)} \rangle_{\text{Weyl}} \sim \frac{\mathbf{1}}{\mathbf{(4\pi\sigma^2)^2}} \mathbf{e^{-r^2/(4\sigma^2)}}$$

   This term, being an entire function in momentum space, satisfies the condition of super-exponential locality (1) as the decay is faster than any power law r^{-k}.

2. **Violation Contribution ($\Delta\rho \sim \mu^{\beta-1}$ with $\beta > 1$):** If the zeta function has a pole with $\text{Re}(\rho) = \beta > 1$, then $\Delta\rho(\mu)$ grows polynomially as $\mu^{\beta-1}$ in the UV limit ($\mu \to \infty$).

The Paley–Wiener Theorem for generalized functions states that exponential decay in coordinate space (locality) is equivalent to the Fourier transform (the propagator/kernel in momentum space) being an entire function of a specific type.

- The Gaussian factor $e^{-\sigma^2 \mu}$ ($e^{-\sigma^2 k^2}$ in momentum space) is an entire function of order 1, which guarantees UV-finiteness and super-exponential locality/causality for $r \gg \sigma$.
- The appearance of $\Delta\rho \sim \mu^{\beta-1}$ for $\beta > 1$ in the UV-limit **violates** this entire function property. The additional contribution $\Delta\langle \hat{C}(r) \rangle$ for $r \gg \sigma$ will acquire an asymptotic tail that is **not** suppressed super-exponentially. This term introduces power-law decay of the form $r^{-(4-2\beta)}$ for large r, immediately violating the strict locality requirement (1).

Conclusion

The geometric requirement that the operator $\hat{C}$ must enforce super-exponential locality (1) is a strong constraint on the spectral distribution. This proof demonstrates that the physical principle of locality places a strict mathematical boundary on the singularities of the spectral zeta function: any pole ρ with $\text{Re}(\rho) > 1$ in $\zeta_\Delta(s)$ is analytically precluded by the geometric structure of the correlator. The three conditions are thus mathematically equivalent: